

Leminar 5

1. Type and stability of linear planar systems
2. Type and stability of the equilibria of nonlinear planar systems using the linearization method

① Type and stability of linear planar systems

consider (1) $\dot{X} = AX$, where $A \in M_2(\mathbb{R})$, $\det A \neq 0$.

Denote by $\lambda_1, \lambda_2 \in \mathbb{C}$ the two eigenvalues of A .

We know that $\det A = \lambda_1 \lambda_2$.

Remarks

- 1) $\det A \neq 0 \Leftrightarrow 0 \in \mathbb{R}^2$ is the only equilibrium point of (1).
- 2) $\det A \neq 0 \Leftrightarrow \lambda_1 \neq 0$ and $\lambda_2 \neq 0$.

Definition 1 We say that the equilibrium $0 \in \mathbb{R}^2$ of (1)

is a NODE when $\lambda_1, \lambda_2 \in \mathbb{R}$ and, either

$$\lambda_1 \leq \lambda_2 < 0 \quad \text{or} \quad 0 < \lambda_1 \leq \lambda_2.$$

SADDLE when $\lambda_1, \lambda_2 \in \mathbb{R}$ and $\lambda_1 < 0 < \lambda_2$.

FOCUS when $\lambda_{1,2} = \alpha \pm i\beta$ with $\alpha \neq 0$, $\beta \neq 0$.

CENTER when $\lambda_{1,2} = \pm i\beta$ with $\beta \neq 0$.

Theorem 1 If $\operatorname{Re}(\lambda_1) < 0$ and $\operatorname{Re}(\lambda_2) < 0$ then the equilibrium $0 \in \mathbb{R}^2$ of $\dot{X} = AX$ is a global attractor.

If $\operatorname{Re}(\lambda_1) > 0$ and $\operatorname{Re}(\lambda_2) > 0$ then the equilibrium $0 \in \mathbb{R}^2$ of $\dot{X} = AX$ is a global repeller.

Definition 2. Let $f \in C^1(\mathbb{R}^2, \mathbb{R}^2)$ and $\eta^* \in \mathbb{R}^2$ s.t. $f(\eta^*) = 0$. We say that the equilibrium point η^* of $\dot{X} = f(X)$ is stable if $\forall \varepsilon > 0 \exists \delta > 0$ s.t. whenever $\|\eta - \eta^*\| < \delta$ we have $\|\varphi(t, \eta) - \eta^*\| < \varepsilon$, $\forall t \in (0, \infty)$. We say that it is unstable if it is not stable.

Theorem 2. Any center is stable.
Any saddle is unstable.

Exercises 1) Decide the type and stability of the equilibrium $0 \in \mathbb{R}^2$ of the linear systems:

a) $\begin{cases} \dot{x} = -y \\ \dot{y} = 5x \end{cases}$

b) $\begin{cases} \dot{x} = -x \\ \dot{y} = 5y \end{cases}$

c) $\begin{cases} \dot{x} = -3x \\ \dot{y} = -2y \end{cases}$

d) $\begin{cases} \dot{x} = x - y \\ \dot{y} = x + y \end{cases}$

e) $\begin{cases} \dot{x} = 4x - 5y \\ \dot{y} = x - 2y \end{cases}$

2) For what values of the real parameter a , the system $\begin{cases} \dot{x} = ax - 5y \\ \dot{y} = x - 2y \end{cases}$ has a center at $0 \in \mathbb{R}^2$?

3) There exists uncoupled linear systems with a center?

Solutions

1) a)
$$\begin{cases} \dot{x} = -y \\ \dot{y} = 5x \end{cases}$$
 The matrix's system is $A = \begin{pmatrix} 0 & -1 \\ 5 & 0 \end{pmatrix}$

The equation of the eigenvalues of A is $\begin{vmatrix} -d & -1 \\ 5 & -d \end{vmatrix} = 0$

$$\Leftrightarrow d^2 + 5 = 0 \Leftrightarrow d_{1,2} = \pm i\sqrt{5}$$

By Def 1, the equilibrium $0 \in \mathbb{R}^2$ of a) is a center;
and by Th 2, is stable.

b)
$$\begin{cases} \dot{x} = -x \\ \dot{y} = 5y \end{cases}$$
 $A = \begin{pmatrix} -1 & 0 \\ 0 & 5 \end{pmatrix}$ is a diagonal matrix $\Rightarrow$

$\Rightarrow d_1 = -1$ and $d_2 = 5$. We have $d_1 < 0$ and $d_2 > 0$.
Then, by Def 1, 0 of b) is a saddle; and, by Th 2,
it is unstable.

c)
$$\begin{cases} \dot{x} = -3x \\ \dot{y} = -2y \end{cases}$$
 $A = \begin{pmatrix} -3 & 0 \\ 0 & -2 \end{pmatrix}$ diagonal $\Rightarrow d_1 = -3$
 $d_2 = -2$

We have $d_1 < 0$ and $d_2 < 0$.
Then, by Def 1, 0 of c) is a node; and, by Th. 1,
is a global attractor.

d) and e) Homework

$$2) \begin{cases} \dot{x} = ax - 5y \\ \dot{y} = x - 2y \end{cases} \quad A = \begin{pmatrix} a & -5 \\ 1 & -2 \end{pmatrix}$$

the equation of the eigenvalues of A is $\lambda^2 - (\text{tr}A)\lambda + \det A = 0$
 $\text{tr}A = a + (-2) = a - 2$, $\det A = a \cdot (-2) - (-5) \cdot 1 = 5 - 2a$

$$(*) \quad \lambda^2 - (a-2)\lambda + (5-2a) = 0$$

We have that, by Def 1, the equilibrium 0 of this system is a center if and only if $\exists \beta \in \mathbb{R}^*$ s.t. $\lambda_{1,2} = \pm i\beta$ are the eigenvalues of A if and only if

$\exists \beta \in \mathbb{R}^*$ s.t. $\lambda_{1,2} = \pm i\beta$ are the roots of $(*)$ if and only if $\lambda_1 + \lambda_2 = a-2$ and $\lambda_1 \lambda_2 = 5-2a \mid \Leftrightarrow$
 $\lambda_{1,2} = \pm i\beta$

$$\Leftrightarrow 0 = a-2 \quad \text{and} \quad (i\beta) \cdot (-i\beta) = 5-2a \quad (\Rightarrow)$$

$$\Leftrightarrow a=2 \quad \text{and} \quad \beta^2 = 5-2 \cdot 2 \quad \Leftrightarrow a=2 \quad \text{and} \quad \beta^2=1$$

Conclusion 0 is a center of this system $\Leftrightarrow a=2$.

3) An uncoupled linear system (planar) has the form
 $\begin{cases} \dot{x} = ax \\ \dot{y} = by \end{cases}$ with $a, b \in \mathbb{R}$. its matrix is $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$

with the eigenvalues $\lambda_1 = a$ and $\lambda_2 = b$.

According to Def 1, since both eigenvalues are real,
 0 can not be a center.

Notice that it can be either a node, or a saddle.

② Stability of the equilibrium points of nonlinear planar systems using the linearization method

Consider (1) $\dot{X} = f(X)$, where $f \in C^1(\mathbb{R}^2, \mathbb{R}^2)$.

Recall that $\eta^* \in \mathbb{R}^2$ is an equilibrium of (1) $\Leftrightarrow f(\eta^*) = 0$.

Exercise. Find the equilibria of $\begin{cases} \dot{x} = x(1-x) \\ \dot{y} = y(3-y) \end{cases}$.

Answer: $(0,0)$, $(1,0)$, $(0,3)$, $(1,3)$.

The linearization method

We consider the linear system

$$(2) \quad \dot{X} = Jf(\eta^*)X,$$

which is called the linearization of (1) around η^* .

Recall: For a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $(x,y) \mapsto \begin{pmatrix} f_1(x,y) \\ f_2(x,y) \end{pmatrix}$, the Jacobian matrix of f computed in (x,y) is

$$Jf(x,y) = \begin{pmatrix} \frac{\partial f_1}{\partial x}(x,y) & \frac{\partial f_1}{\partial y}(x,y) \\ \frac{\partial f_2}{\partial x}(x,y) & \frac{\partial f_2}{\partial y}(x,y) \end{pmatrix}.$$

Definition 3 We say that the equilibrium point η^* of $\dot{X} = f(X)$ is hyperbolic if $\operatorname{Re}(\lambda_1) \neq 0$ and $\operatorname{Re}(\lambda_2) \neq 0$, where $\lambda_{1,2} \in \mathbb{C}$ are the eigenvalues of $Jf(\eta^*)$.

Note: $\operatorname{Re}(\lambda)$ denotes the real part of $\lambda \in \mathbb{C}$.

Theorem 3 Let η^* be a hyperbolic equilibrium point of $\dot{X} = f(X)$. If the equilibrium $0 \in \mathbb{R}^2$ of $\dot{X} = Jf(\eta^*)X$ is an attractor (repeller) then the equilibrium η^* of $\dot{X} = f(X)$ is also an attractor (repeller). If the equilibrium point $0 \in \mathbb{R}^2$ of $\dot{X} = Jf(\eta^*)X$ is a saddle, then the equilibrium point η^* of $\dot{X} = f(X)$ is unstable.

Exercise 4) Study the stability of the equilibria of the system

$$\begin{cases} \dot{x} = x(1-x) \\ \dot{y} = y(3-y) \end{cases}$$

Solution Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(x, y) = \begin{pmatrix} x - x^2 \\ 3y - y^2 \end{pmatrix}$.

We have that $f(x, y) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ has the roots $(0, 0)$, $(1, 0)$, $(0, 3)$, $(1, 3)$. Thus, these 4 points are the equilibria of the given system. We need to compute $Jf(x, y) = \begin{pmatrix} 1-2x & 0 \\ 0 & 3-2y \end{pmatrix}$.

In order to study the stability of the equilibrium $(0, 0)$, we compute $Jf(0, 0) = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$. Its eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 3$, both real and positive.

By Th 1, the eq. $0 \in \mathbb{R}^2$ of $\dot{X} = Jf(0, 0)X$ is a repeller. Thus, by Th 3, the eq. $(0, 0) \in \mathbb{R}^2$ of the given nonlinear system is also a repeller.

In order to study the stability of the equilibrium $(1,0)$, we compute $Jf(1,0) = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}$. Its eigenvalues are

$$\lambda_1 = -1 < 0 \text{ and } \lambda_2 = 3 > 0.$$

By Def. 1, the eq. $(0,0)$ of $\dot{X} = Jf(1,0)X$ is a saddle. Thus, by Th3, the equilibrium $(1,0)$ of the given nonlinear system is unstable.

Homework: study the stability of $(0,3)$ and $(1,3)$.

Important remark In order to study the stability of the equilibria of a second order nonlinear equation (scalar)

$$\ddot{x} = f(x, \dot{x}),$$

we transform it into a planar system with the unknowns x and $y = \dot{x}$. So, the planar system is

$$\begin{cases} \dot{x} = y \\ \dot{y} = f(x, y) \end{cases}.$$

Exercise 5) Study the stability of the equilibria of

$$\ddot{\theta} + \omega^2 \sin \theta = 0 \quad (\text{the idealized equation of a pendulum}).$$

Solution. Denote $x = \theta$ and $y = \dot{\theta}$. The equivalent planar system is

$$\begin{cases} \dot{x} = y \\ \dot{y} = -\omega^2 \sin x \end{cases}.$$

In order to find the equilibria, we must solve

$$\begin{cases} y = 0 \\ \sin x = 0 \end{cases}.$$

Thus, the equilibria are $(k\pi, 0)$, $k \in \mathbb{Z}$.

Note that, physically, they correspond to only 2 positions:
 $(0,0)$ and $(\pi,0)$.

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(x,y) = \begin{pmatrix} y \\ -\omega^2 \sin x \end{pmatrix}$. We have

$$Jf(x,y) = \begin{pmatrix} 0 & 1 \\ -\omega^2 \cos x & 0 \end{pmatrix}$$

In order to study the stability of the equilibrium $(0,0)$,
we compute $Jf(0,0) = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix}$ whose trace is 0, and
determinant is ω^2 .

Thus, the equation of the eigenvalues is $\lambda^2 - 0 \cdot \lambda + \omega^2 = 0$
 $\Leftrightarrow \lambda^2 + \omega^2 = 0 \Leftrightarrow \lambda_{1,2} = \pm i\omega$

Thus, we have $\operatorname{Re}(\lambda_1) = \operatorname{Re}(\lambda_2) = 0$. By Def. 3,
we have that the equilibrium point $(0,0)$ is not hyperbolic.
Hence, the main hypothesis of Th 3 is not fulfilled
and, for the moment, we do not have another result
to study the stability of $(0,0)$. So, we are not able
to decide (for the moment).

In order to ~~aff~~ study the stability of the equil. $(\pi,0)$,
we compute $Jf(\pi,0) = \begin{pmatrix} 0 & 1 \\ \omega^2 & 0 \end{pmatrix}$ whose trace is 0,
and determinant is $-\omega^2$.
 $\lambda^2 - \omega^2 = 0 \Leftrightarrow \lambda_{1,2} = \pm \omega$. Thus, the eigenvalues are
real, with opposite signs. By Def. 1, the equil. $(0,0)$
of $\dot{X} = Jf(\pi,0)X$ is a saddle; and, by Th 3, the
equil. $(\pi,0)$ of our system is unstable.